

CS & IT ENGINEERING



Computer Network - 1

Transport Layer

Lecture No. – 07

By - Abhishek Sir





Recap of Previous Lecture



Topic

TCP Data Transfer





Topics to be Covered



Topic

TCP Flow Control

Topic

TCP Congestion Control

→ TCP Cong. Control
Algo. (V. Imp)



ABOUT ME



Hello, I'm **Abhishek**

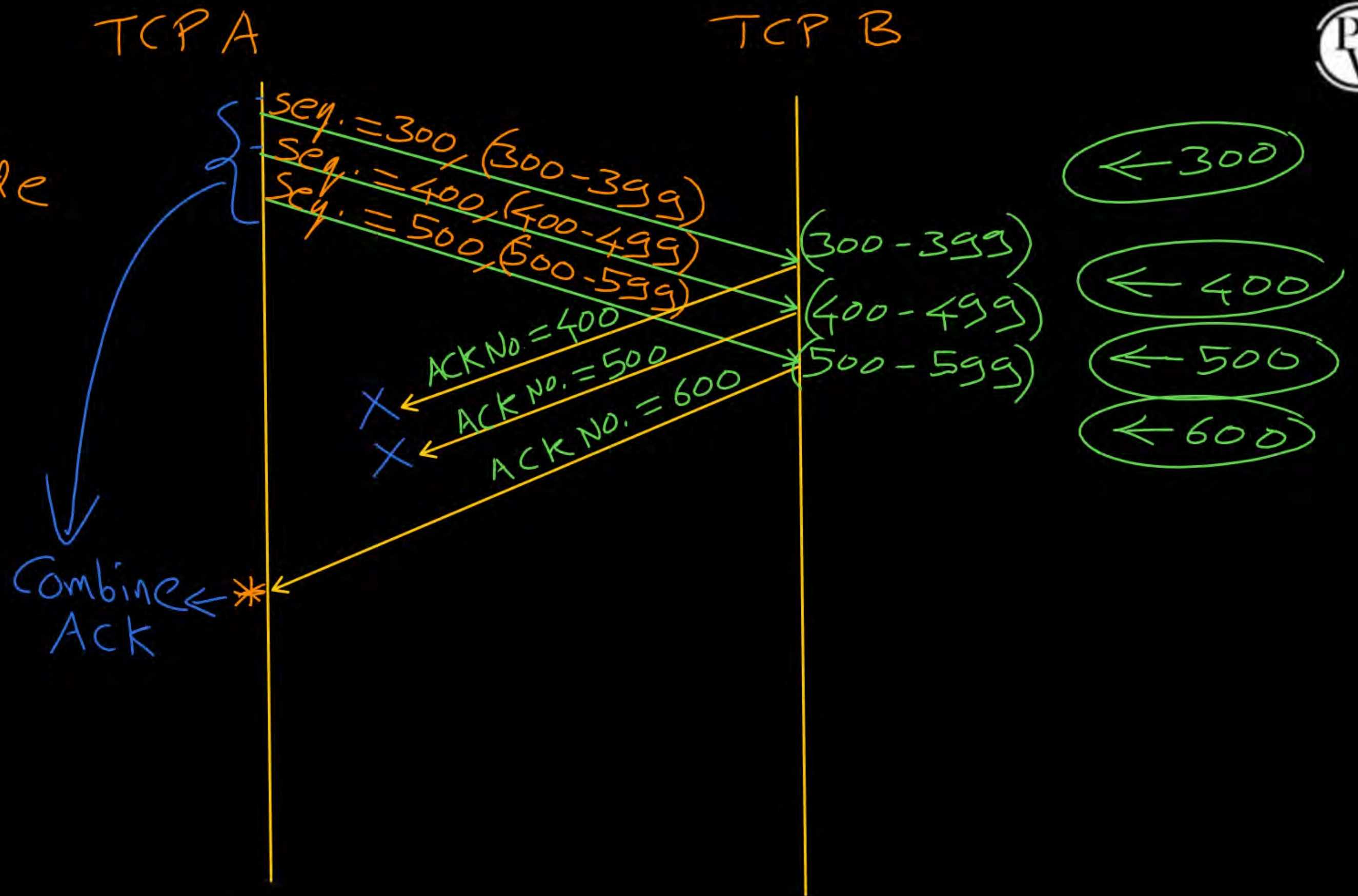
- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



client
ISN=X
=299
MSS=100 byte

CASE I:-



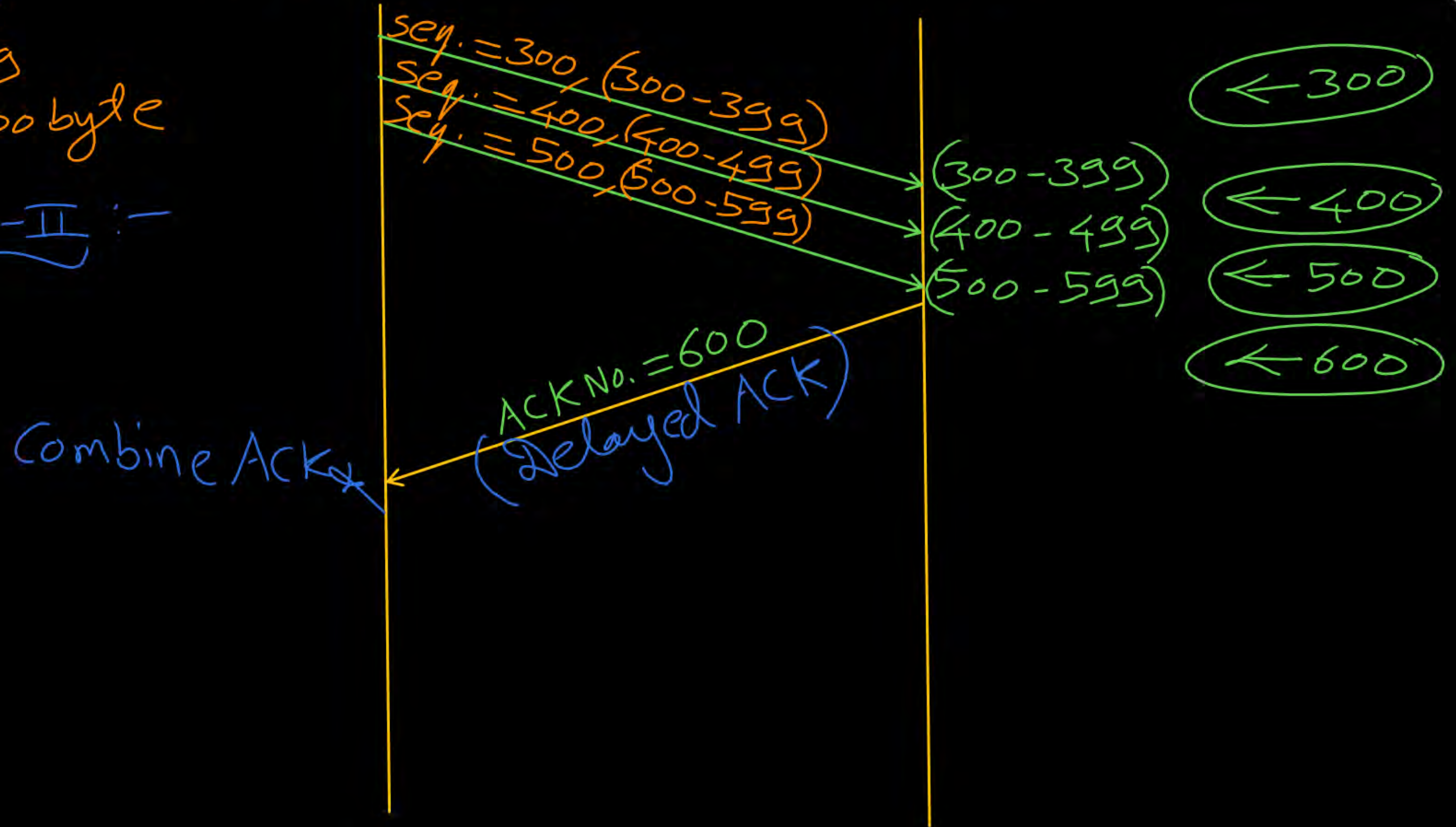


client
ISN = X
= 299
MSS = 100 byte

CASE - II :-

TCP A

TCP B



[MCQ]

[GATE-IT-2007] [1 Mark]



#Q. Consider a TCP connection in a state where there are no outstanding ACKs. The sender sends two segments back to back. The sequence numbers of the first and second segments are 230 and 290 respectively. The first segment was lost, but the second segment was received correctly by the receiver. Let X be the amount of data carried in the first segment (in bytes), and Y be the ACK number sent by the receiver. The values of X and Y (in that order) are :

Outstanding Byte $\rightarrow$ Transmitted Byte, but yet to be acked.

$$X = (290 - 230) = 60 \text{ bytes}$$

$$Y = 230$$

- ☐ A 60 and 290
- ☒ B 230 and 291
- ☐ C 60 and 231
- ☒ D 60 and 230



TCP Sender

TCP Receiver

← 230

Seq_No. = 230,
Payload: [230-289]

Seq_No. = 290,
Payload: [290-]

(Ack_No. = 230)

[290-]



Topic : TCP Buffer



→ Each TCP socket have two buffers :-

1. Send buffer size : MaxSendBuffer ✓
2. Receive buffer size : MaxRcvBuffer ✓

TCP Client Process

P_1

SOCK_STREAM
(TCP Socket)

Send Buffer

(MaxSendBuffer = 1000 Bytes)

Recv Buffer

(MaxRecvBuffer = 1000 Bytes)

⇒ LastByteWritten = P_1 's ISN

LastByteSent = P_1 's ISN

LastByteAcked = P_1 's ISN

RWND =



LastByteRead = P_2 's ISN

NextByteExpected = P_2 's ISN + 1

LastByteRcvd = P_2 's ISN

TCP Server Process

P_2

SOCK_STREAM
(TCP Connection
Socket 1)

Send Buffer

(MaxSendBuffer = 1200 Bytes)

Recv Buffer

(MaxRecvBuffer = 1200 Bytes)

LastByteWritten = P_2 's ISN

LastByteSent = P_2 's ISN

LastByteAcked = P_2 's ISN

RWND



✓
LastByteRead = P_1 's ISN

✓
NextByteExpected = P_1 's ISN + 1

✓
LastByteRcvd = P_1 's ISN



Topic : TCP Buffer


$$\text{LastByteAcked} \leq \text{LastByteSent} \leq \text{LastByteWritten}$$
$$\text{LastByteRead} < \text{NextByteExpected} \leq (\text{LastByteRcvd} + 1)$$



Topic : TCP Buffer



send buffer:—

$$[\text{LastByteWritten} - \text{LastByteAcked}] \leq \text{MaxSendBuffer}$$

Recv Buffer:—

$$[\text{LastByteRcvd} - \text{LastByteRead}] \leq \text{MaxRcvBuffer}$$



Topic : TCP Header



0	16	31
Source Port Number (16 bit)		Destination Port Number (16 bit)
Sequence Number (32 - bits)		
[Acknowledgment Number (32 - bits)]		
HLEN	6 Flag bits	[Window Size (16 bit)]
Checksum (16 bit)		Urgent Pointer (16 bit)
Optional Header		
—//— Payload —//—		



Topic : Window Size



→ Receiver Window Size (16-bit)

→ An advertisement from receiver

→ Define how much data (in bytes) the receiving device can accept

→ Define current available space (in bytes) in receiver's receiving window



Topic : Window Size



→ In every transmitted TCP segment, window size initialized using below equation

$$\text{AdvertisedRcvWindow} = \text{MaxRcvBuffer} - [\text{NextByteExpected} - \text{LastByteRead} - 1]$$

Ex:- MaxRcvBuffer = 1200 byte
NextByteExpected = 600
LastByteRead = 299

$$\begin{aligned} \text{Adv. Recv Window Size} &= [1200 - (600 - 299 - 1)] \\ &= (1200 - 300) = 900 \text{ byte} \end{aligned}$$



Topic : Window Size



→ TCP maintain one more variable regarding 'Send Buffer',
to control the transmission flow

→ RWND : Receiver Advertised Window Size

→ For every received TCP segment,
RWND = AdvertisedRcvWindow



Topic : TCP Flow Control



$$(\text{LastByteSent} - \text{LastByteAcked}) \leq \text{RWND}$$

← Flow Control Equation

$$\text{No. of outstanding bytes} \leq \text{RWND}$$

$$\text{No. of outstanding byte} = (\text{LastByteSent} - \text{LastByteAcked})$$

[Transmitted bytes,
yet to be Acked]



Topic : TCP Flow Control



=> TCP wants to send a segment contains N bytes in payload

if [(LastByteSent - LastByteAked) + N] $\leq$ RWND

→ Fill upto N bytes [(LastByteSent + 1) to (LastByteSent + N)]
in the payload field of segment from 'Send Buffer'

→ Sequence Number = [LastByteSent + 1 ;]

→ ACK Number = [NextByteExpected ;]

→ LastByteSent = LastByteSent + N ;

else

→ Not allowed to send immediately, wait for sometime.



2 mins Summary



Topic

TCP Flow Control ✓

Topic

~~TCP Congestion Control~~



THANK - YOU

